

E.Coli Growth Simulations

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1. Introduction

The study of microbial growth kinetics is a fundamental aspect of biochemical engineering. In a standard batch culture, bacterial growth typically shows four distinct stages: the lag phase, the exponential (log) phase, the stationary phase and the death phase. The objective of this project is to model the growth kinetics of *Escherichia coli* (*E. coli*) under specific operating conditions.

2. Methods and Model

2.1. Assumption

The model is created based on many assumptions. First, it is assumed that the temperature is constant at 37°C throughout the entire process. The growth media is also assumed to be LB broth. The project also based the mathematical model of lag phase on the growth of *E.Coli* cells from a starting LB medium to a more diluted LB growth medium. As this project heavily relied on two mathematical models of Krce et al. (2019) and Fujikawa et al. (2004), all assumptions made in these two papers must also be applied here.

2.2. Baseline model

Most used models are based on the rate logistic (Verhulst) model. However, such a model failed to capture the initial lag phase of cells when encountered a new environment and fail to show the death phase of the population. Thus, to tackle these issues, the project took inspiration from the models proposed by Krce et al. (2019) and Fujikawa et al. (2004).

2.2.1. Krce et al. (2019)

Krce et al. (2019) introduces the following general equation:

$$\frac{d}{dt}B(t) = aN(t)B(t) - bB(t)^2 ; \frac{d}{dt}N(t) = -aN(t)B(t)$$

consists of the two terms of the bacterial concentration $B(t)$ (CFU/ml) and the nutrient concentration $N(t)$ (this is expressed in CFU/ml, interpreted as the concentration of bacteria that could be produced with a given nutrient concentration). Noted that these functions are minute based. The first term describes the interaction between bacteria and nutrients, while the second term is an innovative way to represent the competitive interaction between two cells (the probability of appearance of this reaction must be proportional to the cell concentration square). The two term a and b are interaction strength parameters, found by experimental data.

The parameters are found to varied based on the fraction of initial nutrient concentrations to the nutrient concentration in 100% LB medium, which are converted to x – dilution

factor. Combining with experimental data, the set of equations governing the E.Coli concentration are found to be:

$$q(x) = \frac{x}{x+0.093}; \frac{d}{dt}n(t) = -q(x) * 1.94 \times 10^{-2}b(t)n(t)$$

$$\frac{d}{dt}b(t) = q(x) * [1.94 \times 10^{-2}b(t)n(t) - 1.96 \times 10^{-4}b(t)^2]$$

In this result, the calculations are normalized by initial nutrient concentration, that is $b(t) = B(t)/N0$ and $n(t) = N(t)/N0$. The $q(x)$ is introduced bacterial nutrient uptake efficiency, which is inversely proportional to food availability or dilution factor; bacteria feed highly efficiently when food is scarce but become less efficient when food is abundant.

However, such a model still fails to induce a lag phase, prompting the need of a supporting mathematical model for this initial phase.

2.2.2. Fujikawa et al. (2004)

This research also based its model on the rate logistic (Verhulst) model but modified it to induce a lag phase. The results based on experimental data are as follows:

$$\frac{d}{dt}B(t) = r * B(t) * \left(1 - \frac{B(t)}{B_{\max}}\right) * \left(1 - \frac{B_{\min}}{B(t)}\right)^{0.72}$$

where r is a constant based on temperature. More noticeably, the research assumed a new term $\left(1 - \frac{B_{\min}}{B(t)}\right)^{0.72}$ (or lag term) to recreate the cell lag phase. It is explained that when bacteria concentration B is near $B_{\min}$ in the lag phase, the value of this term is very small, thus making the rate of growth very low. As B increases towards $B_{\max}$, the term is almost 1, and making the cell follow a normal logistic model. In this paper, the term $B_{\min}$ is set to be $(1 - 10^{-6})$ times of Cell Inoculation.

Nevertheless, this model fails to account for the death phase.

2.2.3. The Mathematical Model

The project took the idea of the death phase model of Krce et al. (2019) and the lag term of Fujikawa et al. (2004) to create a final set of differential equation to model E.Coli growth. The equations are as follows:

$$q(x) = \frac{x}{x + 0.093}$$

$$\frac{d}{dt}b(t) = q(x) * [1.94 \times 10^{-2}b(t)n(t) - 1.96 \times 10^{-4}b(t)^2] * \left(1 - \frac{B_{\min}}{B(t)}\right)^{0.72}$$

$$\frac{d}{dt}n(t) = -q(x) * 1.94 \times 10^{-2}b(t)n(t) * (1 - \frac{B_{min}}{B(t)})^{0.72}$$

The model aims to incorporate the ability of the model proposed by Krce et al. (2019) to capture the cell death phase, while integrating the lag term from the Fujikawa et al. (2004) model to address the former model's inability to represent the lag phase.

2.2.4. Matlab Code

Matlab was used to plot the project model and compare with model of the other two papers. Additionally, the code also introduces an interface where parameters such as Initial Inoculation Size (10^8 CFU/ml), Growth Medium dilution factor (compared to 100% LB medium) and Simulation Time (minutes) can be chosen freely.

The Matlab App interface can be shown in **Fig 1**. The full Matlab Code can be found in Appendix. The Run function will plot the Project Model, while the Compare button will plot the two base models for comparison. The Clear button is used to reset the graph.

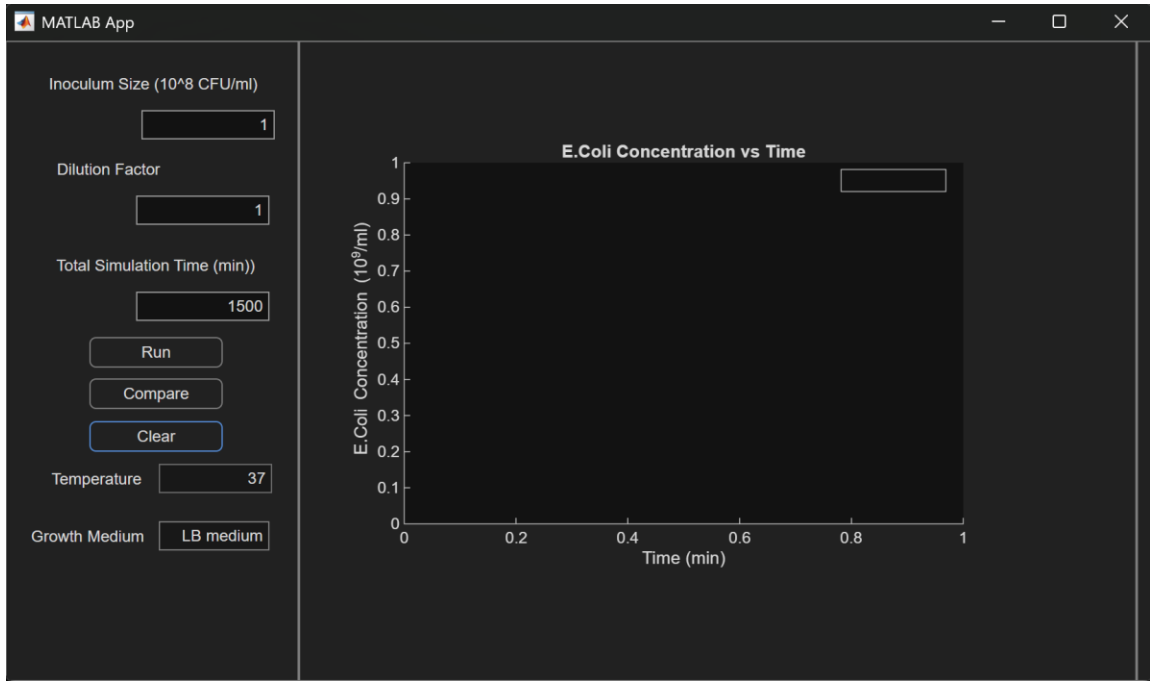


Fig 1: Matlab App interface

3. Results and Discussion

By choosing the basic initial parameters of Inoculum size 1, Dilution Factor of 1 (or 100% LB medium) and Simulation Time of 1500 minutes, the plot of all 3 models can be shown in **Fig 2**.

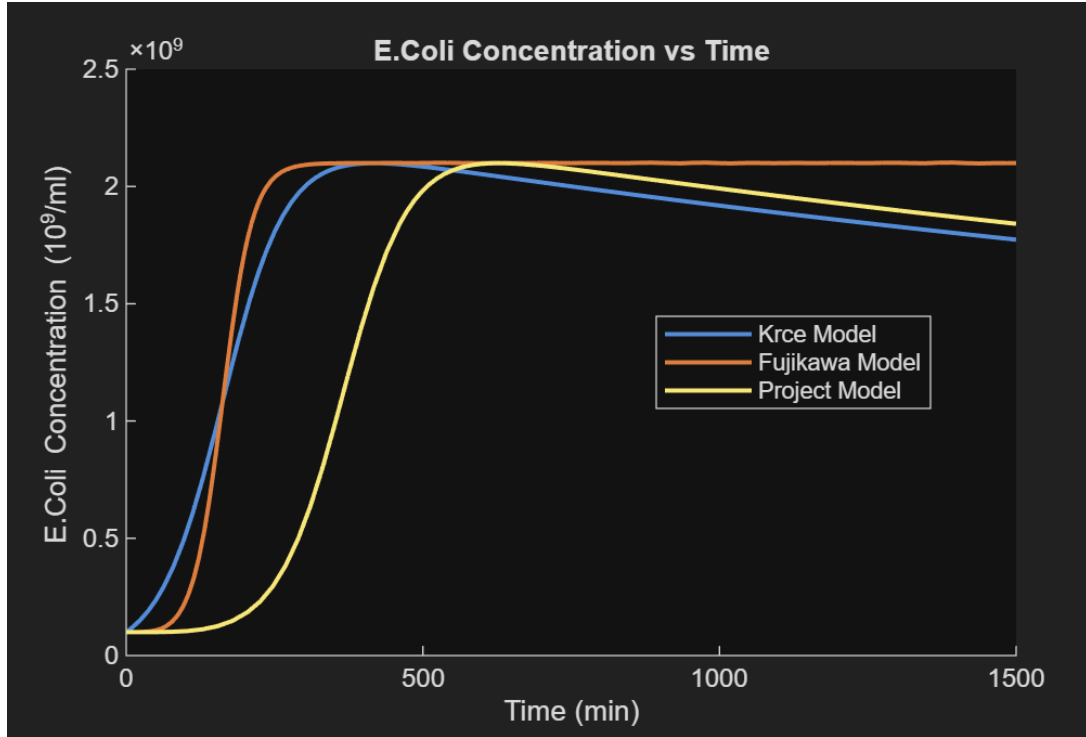


Fig 2: All 3 models under basic initial conditions

The Project Model successfully integrates the advantages of both mathematical models. Specifically, it inherits the ability of Krce's Model to capture the decline in cell concentration during the final phase. This occurs because, at high cell concentrations, the term $\frac{B_{\min}}{B(t)}$ approaches 0 as $B(t)$ becomes very large, causing the lag term to approach 1. Consequently, under these conditions, the Project Model behaves similarly to Krce's Model. This can be seen more clearly when changing the simulation time to 15000 mins in **Fig 3a**. The cell concentration decreases in both the Krce and Project Models, whereas the Fujikawa model shows no decline due to the absence of a term representing cell concentration reduction.

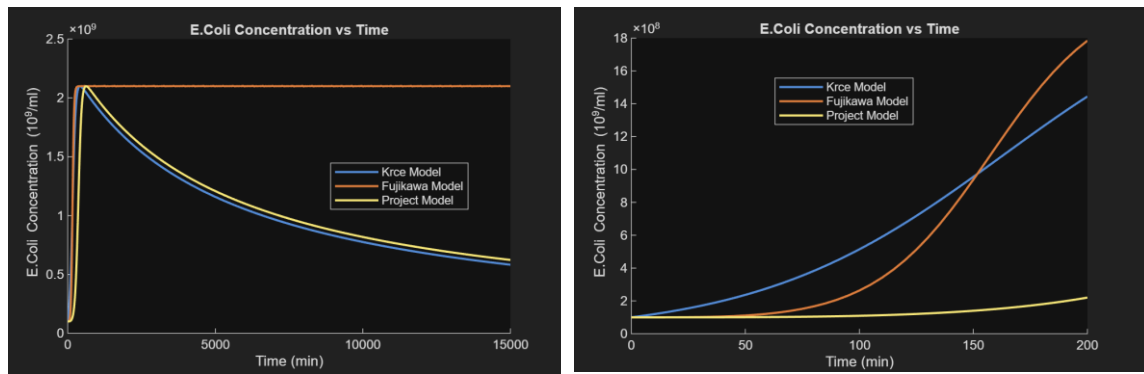


Fig 3: All 3 models under a) large Simulation Time b) low Simulation Time

The lag phase of the Project Model is observed to be significantly longer than that of Fujikawa Model. This is attributed to the Project Model inheriting the initial growth rate from the Krce Model rather than the Fujikawa Model. To be more specific, rather than adopting the growth constant r of 0.0411 min^{-1} , the Project Model uses a smaller initial growth rate of the Krce model, $k_m = 0.0194 \text{ min}^{-1}$, resulting in a prolonged lag phase. This behavior can be seen more clearly when a low simulation time of 200 minutes is used in **Fig 3b**, where both base models have already entered the exponential phase, while the Project Model still indicates that the cells remain in the lag phase.

However, comparison with the base models alone is insufficient to validate the Project Model, as neither model adequately accounts for such an extended lag phase. Krce et al. (2019) did not consider the lag phase, while Fujikawa et al. (2004) only incorporated lag phase effects when using growth media other than LB medium. Therefore, further experimental validation is required to determine whether the prolonged lag phase predicted by the Project Model accurately reflects actual bacterial growth behavior. Furthermore, it should be noted that the lag phase in this model is forced, which may not accurately represent an identical LB-to-LB medium transfer. Therefore, the model should only be considered applicable for simulating nutritional transition shock or changes in growth medium.

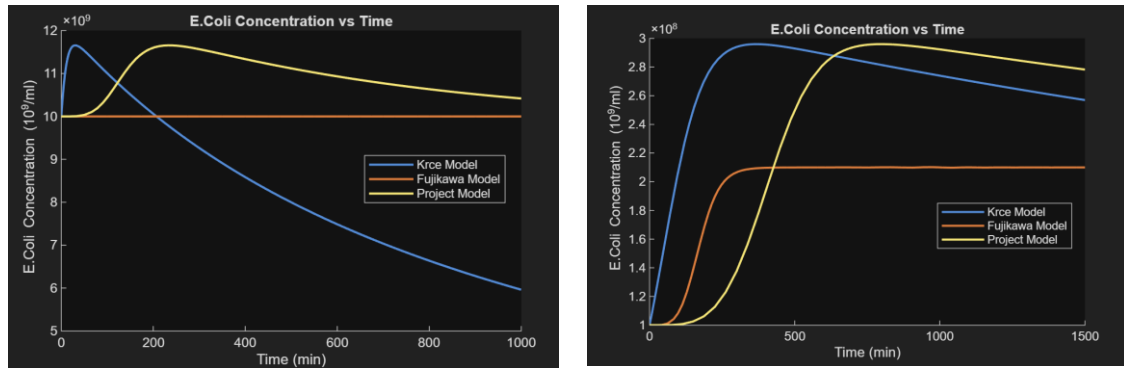


Fig 4: All 3 models under a) high Cell Inoculation b) low Dilution Factors

Despite their similar behavior at low cell inoculation levels, the models show a large discrepancy at high cell inoculation levels, as observed in **Fig 4a** where the cell inoculation is set to 100. Under these conditions, the Fujikawa Model predicts that the cells already reach their maximum population, whereas the Krce et al. model indicates that the cell concentration can still increase to another peak before declining. This discrepancy between the two base models results in the Project Model failing to accurately represent the cell concentration behavior under high inoculation conditions.

Another significant difference can be observed in low dilution factors, such as in **Fig 4b** where the dilution factor is set at 0.1 (10% LB medium). This occurs because the Fujikawa model completely ignores the dilution factor term. Consequently, the Fujikawa model in **Fig 4b** incorrectly predicts that the cells continue to grow at a rate similar to that in a 100% LB medium, whereas the Krce model accounts for the reduced growth rate in a diluted medium.

The difference in maximum cell concentration can also be attributed to the differences in the mathematical formulations of the two models. While the Krce model determines cell concentration based on the balance between nutrient availability and cell concentration, the Fujikawa model imposes a “hard ceiling” through the term $\left(1 - \frac{B(t)}{B_{\max}}\right)$. Nevertheless, the Project Model is still partially able to follow the behavior of the Krce model, as it does not inherit this hard ceiling term.

From such discussions, it is suggested that the Project Model most accurately represents cell growth in low Cell Inoculation and maximum Dilution Factor condition.

4. Conclusions

The Project Model can incorporate both models’ advantages to plot the cell growth over time. However, such accurate representation can only be seen in specific starting conditions, prompting the need for refinement in the future.

5. References

- Fujikawa, H., Kai, A., & Morozumi, S. (2004). A new logistic model for Escherichia coli growth at constant and dynamic temperatures. Food Microbiology, 21(5), 501–509. <https://doi.org/10.1016/j.fm.2004.01.007>
- Krce, L., Šprung, M., Maravić, A., & Aviani, I. (2019). A simple interaction-based E. coli growth model. Physical Biology, 16(6), Article 066005. <https://doi.org/10.1088/1478-3975/ab3d51>

6. Appendix:

6.1. Matlab Code:

```
1. classdef test < matlab.apps.AppBase
2.
3.     % Properties that correspond to app components
4.     properties (Access = public)
5.         UIFigure                matlab.ui.Figure
6.         GridLayout              matlab.ui.container.GridLayout
7.         LeftPanel               matlab.ui.container.Panel
8.         CompareButton           matlab.ui.control.Button
9.         ClearButton             matlab.ui.control.Button
10.        RunButton                matlab.ui.control.Button
11.        GrowthMediumEditField    matlab.ui.control.EditField
12.        GrowthMediumEditFieldLabel matlab.ui.control.Label
13.        TemperatureEditField     matlab.ui.control.NumericEditField
14.        TemperatureEditFieldLabel matlab.ui.control.Label
15.        TotalSimulationTimeminEditField matlab.ui.control.NumericEditField
16.        TotalSimulationTimeminEditFieldLabel matlab.ui.control.Label
17.        DilutionFactorEditField  matlab.ui.control.NumericEditField
18.        DilutionFactorEditFieldLabel matlab.ui.control.Label
19.        InoculumSize108CFUmlEditField matlab.ui.control.NumericEditField
20.        InoculumSize108CFUmlEditFieldLabel matlab.ui.control.Label
21.        CenterPanel              matlab.ui.container.Panel
22.        UIAxes                   matlab.ui.control.UIAxes
23.        RightPanel               matlab.ui.container.Panel
```

```

24.     end
25.
26.     % Properties that correspond to apps with auto-reflow
27.     properties (Access = private)
28.         onePanelWidth = 576;
29.         twoPanelWidth = 768;
30.     end
31.
32.     % Callbacks that handle component events
33.     methods (Access = private)
34.
35.         % Button pushed function: RunButton
36.         function RunButtonPushed(app, event)
37.             % taken from Krce et al., 2019
38.             B0 = app.InoculumSize108CFUmlEditField.Value*(10.^8);
39.             x = app.DilutionFactorEditField.Value;
40.             time = app.TotalSimulationTimeminEditField.Value;
41.             N0 = 2.1*(10.^(9))*x; %the theoretical maximum given the initial
nutrient concentration
42.             km = 1.94*(10.^(-2)); kbb = 1.96*(10.^(-4)); %experimental data
43.             xhalf = 0.093;
44.             qx = x/(x+xhalf);
45.             c = 0.72;
46.             Nmin = (1-1./(10.^6))*B0;
47.             t = [0 time];
48.             function dgdt = dgdtf(t,g,qx,km,kbb,Nmin,c,N0)
49.                 b = g(1); n = g(2);
50.                 Braw = max(1,b*N0);
51.                 lag_factor = max(0,1-Nmin/Braw).^c;
52.                 dgdt(1) = qx*(km*b*n-kbb*(b.^2))*lag_factor;
53.                 dgdt(2,1) = -qx*km*b*n*lag_factor;
54.             end
55.
56.             hold(app.UIAxes, 'on')
57.             fg = @(t,g) dgdtf(t,g,qx,km,kbb,Nmin,c,N0);
58.             [t3,g] = ode45(fg,t,[B0/N0 1]);
59.             C = g(:,1)*N0;
60.             plot(app.UIAxes,t3,C,'y','LineWidth', 1.5, 'DisplayName', 'Project
Model')
61.             hold(app.UIAxes, 'off')
62.             legend(app.UIAxes, 'show', 'Location', 'best')
63.
64.         end
65.
66.         % Value changed function: TemperatureEditField
67.         function TemperatureEditFieldValueChanged(app, event)
68.             app.TemperatureEditField.Value = 37;
69.         end
70.
71.         % Value changed function: GrowthMediumEditField
72.         function GrowthMediumEditFieldValueChanged(app, event)
73.             app.GrowthMediumEditField.Value = 'LB medium';
74.         end
75.
76.         % Callback function

```

```

77.     function umaxEditFieldValueChanged(app, event)
78.         app.umaxEditField.Value = 0; %change to real umax
79.     end
80.
81.     % Callback function
82.     function KsEditFieldValueChanged(app, event)
83.         app.KsEditField.Value = 0; %change to real Ks
84.     end
85.
86.     % Button down function: UIAxes
87.     function UIAxesButtonDown(app, event)
88.
89.     end
90.
91.     % Button pushed function: ClearButton
92.     function ClearButtonPushed(app, event)
93.         cla(app.UIAxes);
94.     end
95.
96.     % Button pushed function: CompareButton
97.     function CompareButtonPushed(app, event)
98.         B0 = app.InoculumSize108CFUmlEditField.Value*(10.^8);
99.         x = app.DilutionFactorEditField.Value;
100.         time = app.TotalSimulationTimeminEditField.Value;
101.         N0 = 2.1*(10.^(9))*x; %the theoretical maximum given the
    initial nutrient concentration
102.         km = 1.94*(10.^(-2)); kbb = 1.96*(10.^(-4));
103.         xhalf = 0.093;
104.         qx = x/(x+xhalf);
105.
106.         function dydt = dydtf(t,y,qx,km,kbb)
107.             b = y(1); n = y(2);
108.             dydt(1) = qx*(km*b*n-kbb*(b.^2));
109.             dydt(2,1) = -qx*km*b*n;
110.         end
111.
112.         hold(app.UIAxes, 'on')
113.         f = @(t,y) dydtf(t,y,qx,km,kbb);
114.         t = [0 time];
115.         [t1,y] = ode45(f,t,[B0/N0 1]);
116.         B = y(:,1)*N0;
117.         plot(app.UIAxes,t1,B,'Color', [0 0.4470 0.7410],'LineWidth',
    1.5, 'DisplayName', 'Krce Model')
118.
119.         hold(app.UIAxes, 'on')
120.         r = exp(21-6230./(37+273))/60;
121.         c = 0.72;
122.         Nmin = (1-1./(10.^6))*B0;
123.
124.         function dzdt = dzdtf(t,z,r,c,Nmin)
125.             lag_factor = max(0,1-Nmin/z).^c;
126.             dzdt = r*z*(1-z/N0)*lag_factor;
127.         end
128.
129.         fz = @(t,z) dzdtf(t,z,r,c,Nmin);

```



```

130.         [t2,z] = ode45(fz,t,B0);
131.         plot(app.UIAxes,t2,z,'Color',[1 0.5 0],'LineWidth',1.5,
'DisplayName','Fujikawa Model')
132.         hold(app.UIAxes,'off')
133.         legend(app.UIAxes,'show','Location','best')
134.
135.     end
136.
137.     % Changes arrangement of the app based on UIFigure width
138.     function updateAppLayout(app, event)
139.         currentFigureWidth = app.UIFigure.Position(3);
140.         if(currentFigureWidth <= app.onePanelWidth)
141.             % Change to a 3x1 grid
142.             app.GridLayout.RowHeight = {480, 480, 480};
143.             app.GridLayout.ColumnWidth = {'1x'};
144.             app.CenterPanel.Layout.Row = 1;
145.             app.CenterPanel.Layout.Column = 1;
146.             app.LeftPanel.Layout.Row = 2;
147.             app.LeftPanel.Layout.Column = 1;
148.             app.RightPanel.Layout.Row = 3;
149.             app.RightPanel.Layout.Column = 1;
150.         elseif (currentFigureWidth > app.onePanelWidth &&
currentFigureWidth <= app.twoPanelWidth)
151.             % Change to a 2x2 grid
152.             app.GridLayout.RowHeight = {480, 480};
153.             app.GridLayout.ColumnWidth = {'1x', '1x'};
154.             app.CenterPanel.Layout.Row = 1;
155.             app.CenterPanel.Layout.Column = [1,2];
156.             app.LeftPanel.Layout.Row = 2;
157.             app.LeftPanel.Layout.Column = 1;
158.             app.RightPanel.Layout.Row = 2;
159.             app.RightPanel.Layout.Column = 2;
160.         else
161.             % Change to a 1x3 grid
162.             app.GridLayout.RowHeight = {'1x'};
163.             app.GridLayout.ColumnWidth = {220, '1x', 12};
164.             app.LeftPanel.Layout.Row = 1;
165.             app.LeftPanel.Layout.Column = 1;
166.             app.CenterPanel.Layout.Row = 1;
167.             app.CenterPanel.Layout.Column = 2;
168.             app.RightPanel.Layout.Row = 1;
169.             app.RightPanel.Layout.Column = 3;
170.         end
171.     end
172. end
173.
174. % Component initialization
175. methods (Access = private)
176.
177.     % Create UIFigure and components
178.     function createComponents(app)
179.
180.         % Create UIFigure and hide until all components are created
181.         app.UIFigure = uifigure('Visible','off');
182.         app.UIFigure.AutoResizeChildren = 'off';

```

```

183.         app.UIFigure.Position = [100 100 860 480];
184.         app.UIFigure.Name = 'MATLAB App';
185.         app.UIFigure.SizeChangedFcn = createCallbackFcn(app,
            @updateAppLayout, true);
186.
187.         % Create GridLayout
188.         app.GridLayout = uigridlayout(app.UIFigure);
189.         app.GridLayout.ColumnWidth = {220, '1x', 12};
190.         app.GridLayout.RowHeight = {'1x'};
191.         app.GridLayout.ColumnSpacing = 0;
192.         app.GridLayout.RowSpacing = 0;
193.         app.GridLayout.Padding = [0 0 0 0];
194.         app.GridLayout.Scrollable = 'on';
195.
196.         % Create LeftPanel
197.         app.LeftPanel = uipanel(app.GridLayout);
198.         app.LeftPanel.Layout.Row = 1;
199.         app.LeftPanel.Layout.Column = 1;
200.
201.         % Create InoculumSize108CFUmlEditFieldLabel
202.         app.InoculumSize108CFUmlEditFieldLabel =
            uilabel(app.LeftPanel);
203.         app.InoculumSize108CFUmlEditFieldLabel.HorizontalAlignment =
            'right';
204.         app.InoculumSize108CFUmlEditFieldLabel.Position = [28 437
            161 22];
205.         app.InoculumSize108CFUmlEditFieldLabel.Text = 'Inoculum Size
            (10^8 CFU/ml)';
206.
207.         % Create InoculumSize108CFUmlEditField
208.         app.InoculumSize108CFUmlEditField =
            uieditfield(app.LeftPanel, 'numeric');
209.         app.InoculumSize108CFUmlEditField.Position = [102 406 100
            22];
210.
211.         % Create DilutionFactorEditFieldLabel
212.         app.DilutionFactorEditFieldLabel = uilabel(app.LeftPanel);
213.         app.DilutionFactorEditFieldLabel.HorizontalAlignment =
            'right';
214.         app.DilutionFactorEditFieldLabel.Position = [34 373 82 22];
215.         app.DilutionFactorEditFieldLabel.Text = 'Dilution Factor';
216.
217.         % Create DilutionFactorEditField
218.         app.DilutionFactorEditField = uieditfield(app.LeftPanel,
            'numeric');
219.         app.DilutionFactorEditField.Position = [98 342 100 22];
220.         app.DilutionFactorEditField.Value = 1;
221.
222.         % Create TotalSimulationTimeminEditFieldLabel
223.         app.TotalSimulationTimeminEditFieldLabel =
            uilabel(app.LeftPanel);
224.         app.TotalSimulationTimeminEditFieldLabel.HorizontalAlignment
            = 'right';
225.         app.TotalSimulationTimeminEditFieldLabel.Position = [33 301
            154 22];

```

```

226.         app.TotalSimulationTimeminEditFieldLabel.Text = 'Total
Simulation Time (min)';
227.
228.         % Create TotalSimulationTimeminEditField
229.         app.TotalSimulationTimeminEditField =
uieditfield(app.LeftPanel, 'numeric');
230.         app.TotalSimulationTimeminEditField.Position = [98 270 100
22];
231.         app.TotalSimulationTimeminEditField.Value = 1500;
232.
233.         % Create TemperatureEditFieldLabel
234.         app.TemperatureEditFieldLabel = uilabel(app.LeftPanel);
235.         app.TemperatureEditFieldLabel.HorizontalAlignment = 'right';
236.         app.TemperatureEditFieldLabel.Position = [30 141 72 22];
237.         app.TemperatureEditFieldLabel.Text = 'Temperature';
238.
239.         % Create TemperatureEditField
240.         app.TemperatureEditField = uieditfield(app.LeftPanel,
'numeric');
241.         app.TemperatureEditField.ValueChangedFcn =
createCallbackFcn(app, @TemperatureEditFieldValueChanged, true);
242.         app.TemperatureEditField.Editable = 'off';
243.         app.TemperatureEditField.Position = [115 141 85 22];
244.         app.TemperatureEditField.Value = 37;
245.
246.         % Create GrowthMediumEditFieldLabel
247.         app.GrowthMediumEditFieldLabel = uilabel(app.LeftPanel);
248.         app.GrowthMediumEditFieldLabel.HorizontalAlignment =
'right';
249.         app.GrowthMediumEditFieldLabel.Position = [16 98 88 22];
250.         app.GrowthMediumEditFieldLabel.Text = 'Growth Medium';
251.
252.         % Create GrowthMediumEditField
253.         app.GrowthMediumEditField = uieditfield(app.LeftPanel,
'text');
254.         app.GrowthMediumEditField.ValueChangedFcn =
createCallbackFcn(app, @GrowthMediumEditFieldValueChanged, true);
255.         app.GrowthMediumEditField.Editable = 'off';
256.         app.GrowthMediumEditField.HorizontalAlignment = 'right';
257.         app.GrowthMediumEditField.Position = [115 98 83 22];
258.         app.GrowthMediumEditField.Value = 'LB medium';
259.
260.         % Create RunButton
261.         app.RunButton = uibutton(app.LeftPanel, 'push');
262.         app.RunButton.ButtonPushedFcn = createCallbackFcn(app,
@RunButtonPushed, true);
263.         app.RunButton.Position = [63 235 100 23];
264.         app.RunButton.Text = 'Run';
265.
266.         % Create ClearButton
267.         app.ClearButton = uibutton(app.LeftPanel, 'push');
268.         app.ClearButton.ButtonPushedFcn = createCallbackFcn(app,
@ClearButtonPushed, true);
269.         app.ClearButton.Position = [63 172 100 23];
270.         app.ClearButton.Text = 'Clear';

```

```

271.
272.         % Create CompareButton
273.         app.CompareButton = uibutton(app.LeftPanel, 'push');
274.         app.CompareButton.ButtonPushedFcn = createCallbackFcn(app,
    @CompareButtonPushed, true);
275.         app.CompareButton.Position = [63 204 100 23];
276.         app.CompareButton.Text = 'Compare';
277.
278.         % Create CenterPanel
279.         app.CenterPanel = uipanel(app.GridLayout);
280.         app.CenterPanel.Layout.Row = 1;
281.         app.CenterPanel.Layout.Column = 2;
282.
283.         % Create UIAxes
284.         app.UIAxes = uiaxes(app.CenterPanel);
285.         title(app.UIAxes, 'E.Coli Concentration vs Time')
286.         xlabel(app.UIAxes, 'Time (min)')
287.         ylabel(app.UIAxes, 'E.Coli Concentration (10^9/ml)')
288.         zlabel(app.UIAxes, 'Z')
289.         app.UIAxes.ButtonDownFcn = createCallbackFcn(app,
    @UIAxesButtonDown, true);
290.         app.UIAxes.Position = [38 85 465 322];
291.
292.         % Create RightPanel
293.         app.RightPanel = uipanel(app.GridLayout);
294.         app.RightPanel.Layout.Row = 1;
295.         app.RightPanel.Layout.Column = 3;
296.
297.         % Show the figure after all components are created
298.         app.UIFigure.Visible = 'on';
299.     end
300. end
301.
302. % App creation and deletion
303. methods (Access = public)
304.
305.     % Construct app
306.     function app = test
307.
308.         % Create UIFigure and components
309.         createComponents(app)
310.
311.         % Register the app with App Designer
312.         registerApp(app, app.UIFigure)
313.
314.         if nargin == 0
315.             clear app
316.         end
317.     end
318.
319. % Code that executes before app deletion
320. function delete(app)
321.
322.     % Delete UIFigure when app is deleted
323.     delete(app.UIFigure)

```

```
324.           end
325.       end
326.   end
```